WEST BENGAL STATE UNIVERSITY

B.Sc. Major 2nd Semester Examination, 2024

CMSDSC202T-COMPUTER SCIENCE (MAJOR)

Time Allotted: 2 Hours | Full Marks: 50

The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.

1. Answer any *five* questions from the following: 2×5 = 10

(a) Explain ADT of stack.

(b) Write the motivation of sparse matrices.

(c) Define a node of a singly linked list in C++.

(d) With a neat diagram represent 4 elements (21, 30, 12, 11) in circular linked list.

(e) What do you mean by external path length?

(f) What is Threaded binary tree?

(g) Evaluate the postfix expression:

2 3 4 * + 5 – 9 /

(h) What do you mean by collision in hashing?

Answer any *five* questions from the following 8×5 = 40

2. (a) Write an algorithm to find the inorder successor of an element in a BST. 4

(b) Find the address of a [5][6] using (i) row major (ii) column major representation, where base address of the array a [–10 : 10, –10 : 10] is 266 and word size = 4. 2+2

3. Write an algorithm to convert any infix expression to prefix expression. Using this algorithm convert the infix expression to prefix expression. (Showing the content of stack in every step). 4+4

$$((a+b)*c-d)/(e\uparrow f-g*h)$$

4. (a) What are the limitations of hashing? 2

(b) Write an algorithm to explain shell sort. 3

(c) What do you mean by internal and external sorting? 3

5. (a) When link representation of a tree is disadvantageous than array representation? 2

(b) List out the steps involved in deleting a node from a binary search tree. 3

(c) Construct an AVL tree for the following data: 3

21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3, 7

6. (a) Write a C++ program to search an element using binary search. 4

(b) Trace the above program to search 23 in the following elements: 4

12, 15, 18, 20, 22, 36, 39, 40, 46

7. (a) Write C++ program to implement insertion sort. 4

(b) Give comparison of searching methods. 4

8. (a) Write a program in C++ to create a single linked list of *n* nodes and reverse the same. 6

(b) Define Priority queues. 2

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Br

```
void (int n, int arr[n], int )
    mid = ~~arr[n/2]~~ arg target )
for(int i=0; i<n; i++)
    int mid = n/2;
    ~~while (mid == arr[n])~~
    if (mid == arr[n/2]
    else if (mid ==
```

```
int main()
{
    int n;
    cout << "Enter ;"
    cin >> n;
    int ~~arr[n]~~ arr;
    for (int i=0, i<n; i++)
        cout << "Enter the ele
        cin >> arr[n];
    int target;
    cout << " Enter the tar
    cin >> " target
    cout << Binar (n, arr[n],
        target)
```

2, 4, 6, 8, 10

node

[n]